NHI NGUYEN

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INTERESTS

Explanation faithfulness, Metacognition in LLMs and agents, Trustworthy AI evaluation, Causal inference

EDUCATION

Courant Institute, New York University — New York, NY
PhD Candidate in Computer Science — *Aug 2023 - Present*
Advisor: Rajesh Ranganath

Yale University — New Haven, CT
BS/MS in Computer Science, BS in Applied Mathematics, Summa Cum Laude — *Aug 2019 - May 2023*
Honors in Computer Science, Honors in Applied Mathematics

PUBLICATIONS

Conference

- What LLMs Explain Is Not What They Believe: Evaluating Explanation Sufficiency Under Models' Own Input Beliefs. **Nhi Nguyen***, Shauli Ravfogel, Rajesh Ranganath. *ICML 2026*
- FIMP: Foundation Model-Informed Message Passing for Graph Neural Networks. Syed Rizvi*, Nazreen Pallikkavaliyaveetil*, David Zhang, Zhuoyang Lyu, **Nhi Nguyen**, Harry Lyu, Benjamin Christensen, Josue Caro, Antonio Fonseca, Emanuele Zappala, Maryam Bagherian, Christopher Averill, Chadi Abdallah, Amin Karbasi, Rex Ying, Maria Brbic, Rahul Dhodapkar, David van Dijk. *TMLR, 2026*
- Explanations that reveal all through the definition of encoding. Aahlad Puli*, **Nhi Nguyen***, Rajesh Ranganath. *NeurIPS, 2024*

Workshop/Preprint

- Do LLMs Lie About What They Use? Benchmark for Metacognitive Truthfulness in Large Language Models. **Nhi Nguyen***, Shauli Ravfogel, Rajesh Ranganath. *ICML 2025*
- A Flow Artist for Embedding High -Dimensional Cellular Data within Vector Fields. Kincaid MacDonald*, Dhananjay Bhaskar*, Guy Thampakkul, **Nhi Nguyen**, Joia Zhang, Tesfa Asmara, Michael Perlmutter, Ian Adelstein, Smita Krishnaswamy. *MLSP, 2023*
- Black Box Causal Inference: Effect Estimation via Meta Prediction. Lucius EJ Bynum, Aahlad Manas Puli, Diego Herrero-Quevedo, **Nhi Nguyen**, Carlos Fernandez-Granda, Kyunghyun Cho, Rajesh Ranganath. *arXiv:2503.05985*

RESEARCH EXPERIENCE

Capital One, First Party Fraud Team — McLean, VA
Data Science Intern — *Jun 2026 - Aug 2026*

- Developed and scaled a graph-based fraud model using Label Propagation and edge weighting to detect organized rings 3 months in advance, mitigating $71M in risk and $474K in losses
- Designed and implemented a modular pipeline and 3-tier evaluation framework (community, statistical, business metrics) to accelerate iteration and uncover new fraud signals from customer connections

Character AI, Post-training Team — Redwood City, CA
Research Intern — *Jun 2025 - Aug 2025*

- Trained an offline reward model on user preference data over entertainment chat model responses, achieving a 15% AUROC improvement to support reinforcement learning and inference-time reranking

VinAI, Residency Program Hanoi, Viet Nam
Research Intern *Jun 2023 - Aug 2023*
- Developed a real-time drunk-driving detection model using features engineered from in-car cameras and sensors, achieving 80% accuracy

Van Dijk Lab at Yale New Haven, CT
Undergraduate Thesis: Attention as Message Passing for Graph Neural Networks *Sep 2022 - May 2023*
Advisor: David van Dijk
- Developed a GNN layer using multi-headed attention to model interpretable feature-level interactions, and designed a cyclic simulation dataset to evaluate interaction recovery; resulted in preprint AMPNet

Summer Undergraduate Math Research at Yale New Haven, CT
Project: Embedding High-Dimensional Cellular Data within Vector Fields *Jun 2022 - Sep 2022*
Advisor: Smita Krishnaswamy
- Developed a neural network that jointly embeds high-dimensional data and its underlying vector field, improving visualization of dynamic structure in single-cell RNA velocity data; published at MLSP 2023

Tietjen Lab, Team Next Gen Research Assistantship New Haven, CT
Project: Cluster protein activities in donated kidneys *Nov 2021 - Jun 2022*
Advisor: Gregory Tietjen
- Studied methods for clustering proteins by shared Gene Ontology annotations and expression activity in donated kidneys, supporting downstream assessment of graft viability

Tobin Undergraduate Research Assistantship New Haven, CT
Project: Order bias and alternative formats in soccer penalty shootouts *Feb 2021 - Aug 2021*
Advisor: Nils Rudi
- Analyzed first-shooter and home-field advantage in penalty shootouts using Bayesian statistical modeling to quantify the bias reduction of alternative shootout orders (AABB vs. standard ABAB)

AWARDS & FELLOWSHIPS

- ICML 2026 Gold Reviewer Award 2026
- DeepMind Scholarship 2023
- MacCracken Doctoral Fellowship, NYU Graduate School of Arts and Sciences 2023
- Russell Henry Chittenden Prize for senior majoring in the natural sciences 2023
- Trumbull College Student Marshals, Trumbull College 2023
- Phi Beta Kappa Junior Inductee 2022

SKILLS

- *Coding:* Python, PyTorch, Hugging Face Transformers, Scikit-learn, Spark, Pandas
- *Workflow:* Multi-GPU training, HPC/SLURM, Weights & Biases, Git, GitHub

REVIEWER

NeurIPS 2025-2026, ICLR 2026, ICML 2026, AAAI 2026, TMLR 2026

GRADUATE TEACHING ASSISTANT

New York University New York, NY
- *CSCI-GA-2565: Machine Learning.* Rajesh Ranganath *SP 2026*

Yale University New Haven, CT
- *Computer Science:* Deep Learning Theory and Applications; Natural Language Processing *SP 2023*
- *Statistics & Data Science:* Stochastic Processes; Optimization and Computation; Data Exploration and Analysis; Probability and Statistics *2020 - 2023*